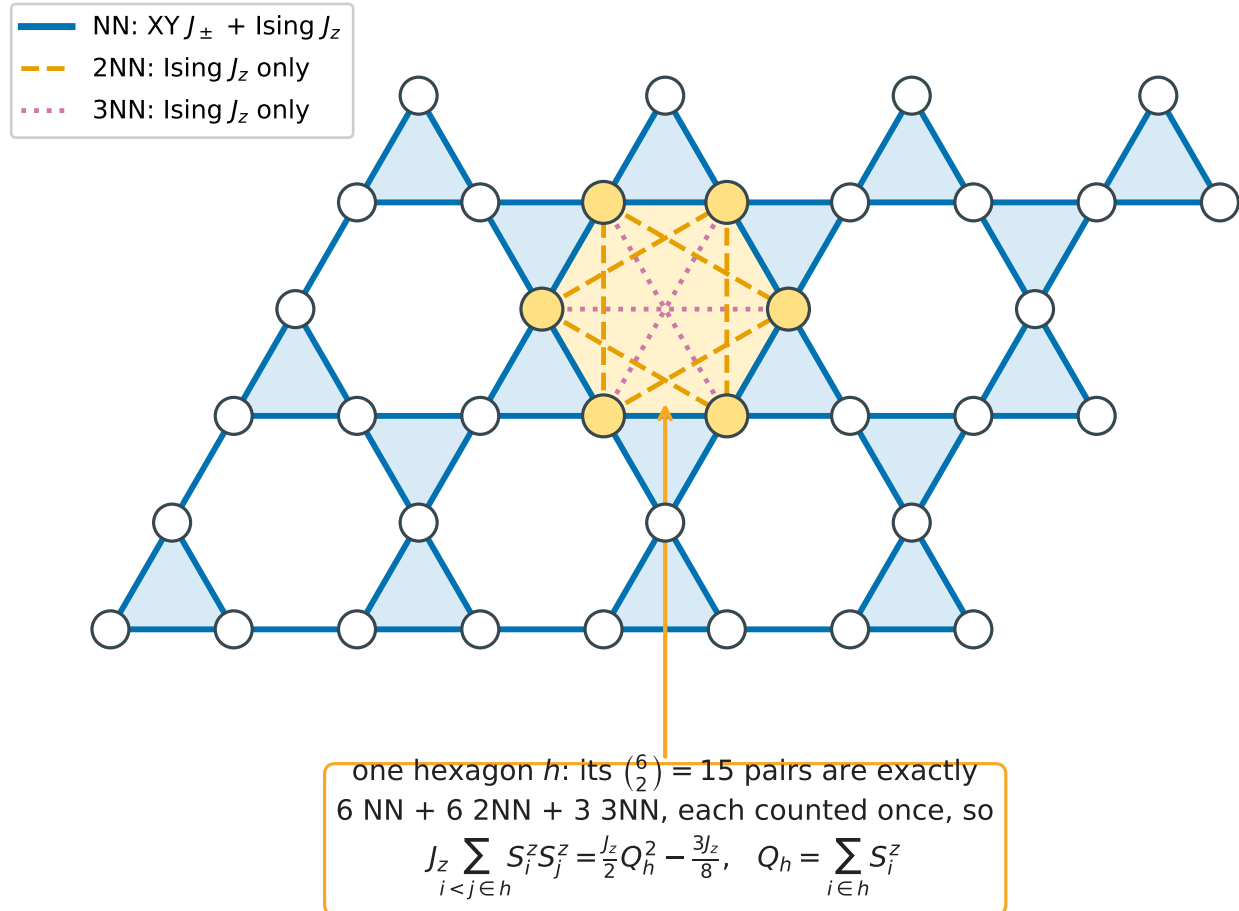
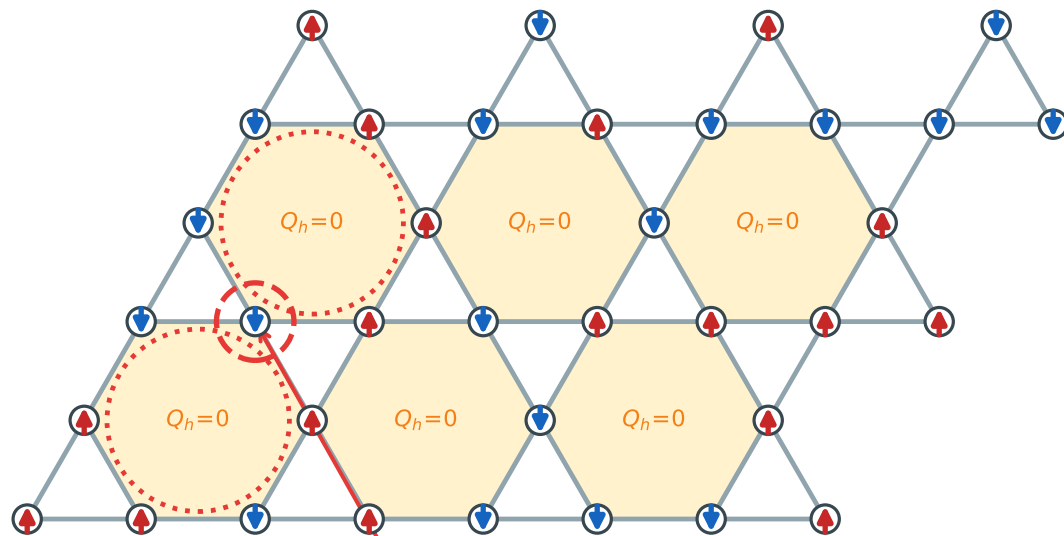


The BFG (Balents-Fisher-Girvin) easy-axis kagome model

(a) kagome bonds — the Ising shells build hexagon charges



(c) ground states at $J_{\pm} = 0$: $Q_h = 0$ (3 $\uparrow$ / 3 $\downarrow$) on every hexagon — extensively degenerate; the constraint is a Z_2 Gauss law



flip this one spin and the two hexagons sharing it go to $Q_h = \pm 1$: a pair of gapped Z_2 charges (spinons), cost $\sim J_z$

(b) the model, and where it lives in the code

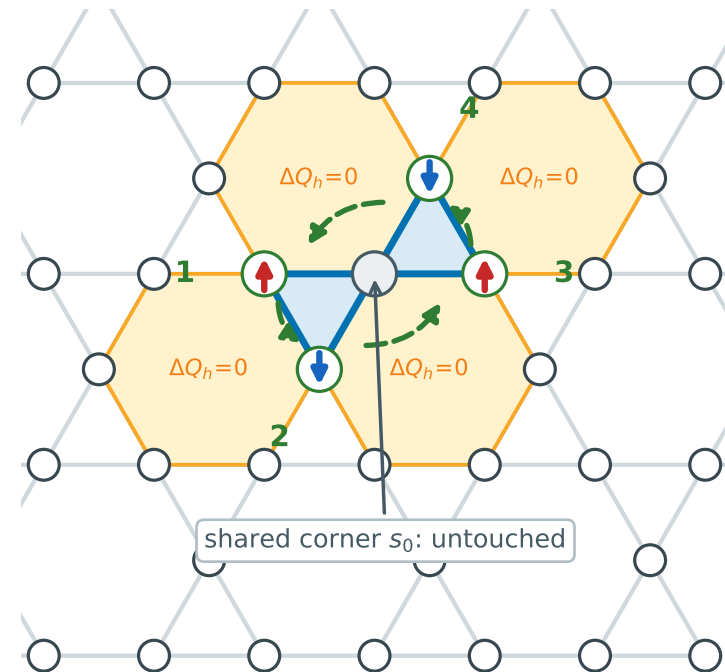
$$H = -J_{\pm} \sum_{\langle ij \rangle_{\text{NN}}} (S_i^+ S_j^- + S_i^- S_j^+) + J_z \sum_{(ij) \in \{\text{NN}, 2\text{NN}, 3\text{NN}\}} S_i^z S_j^z$$

$$= -J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + \text{h.c.}) + \frac{J_z}{2} \sum_{h \in \text{hexagons}} Q_h^2 + \text{const}, \quad Q_h = \sum_{i \in h} S_i^z$$

easy-axis limit $J_{\pm} \ll J_z$ · BFG spin-liquid regime at $J_{\pm} > 0, J_z > 0$

bond shells	helper_kagome_bfg.generate_kagome_cluster NN_BONDS / NN2_BONDS / NN3_BONDS → edges, edges_2nn, edges_3nn
operator, in memory	run_bfg_ground_state.build_bfg_operator op.add_two_body(OP_SPLUS, i, OP_SMINUS, j, -Jpm) # + h.c. op.add_two_body(OP_SZ, i, OP_SZ, j, Jzz) # NN/2NN/3NN
operator, on disk	helper_kagome_bfg.prepare_hamiltonian_parameters → InterAll.dat rows: 2 i 2 j Jzz 0 0 i 1 j -Jpm 0 1 i 0 j -Jpm 0 op codes: 0 = S+, 1 = S-, 2 = Sz
solve	U(1): fixed n_up = N/2 sector; translations Z_dim1 x Z_dim2 one diagonalisation per momentum sector GSD = 4 on the torus is the Z2 topological-order check
observables	compute_bfg_order_parameters.find_bowties .compute_bowtie_resonance → P_bt plus spin structure factor, bond nematicity, chirality

(d) the bow-tie flip — the move that survives the constraint



$$P_{\text{bt}} = \langle S_1^+ S_2^- S_3^+ S_4^- + \text{h.c.} \rangle : |\uparrow \downarrow \uparrow \downarrow \rangle \leftrightarrow |\downarrow \uparrow \downarrow \uparrow \rangle$$

Two triangles sharing ONE corner. The flip acts on the four outer corners and leaves the shared corner alone; the ring alternates a triangle edge (NN) and a hop across the corner (2NN). Consecutive corners sit in a common hexagon, so each of the four hexagons gets one S^+ and one S^- and stays at $Q_h = 0$ — this is the leading move that keeps the state inside the ground-state manifold, generated at order $J_{\pm}^2 / J_z$.